METRIC
FF-B-171/36
30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, DOUBLE ROW, ANGULAR CONTACT, FILLING SLOT, VERTEX OF CONTACT ANGLES OUTSIDE BEARING, ANSI/AFBMA DIMENSION SERIES 32.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

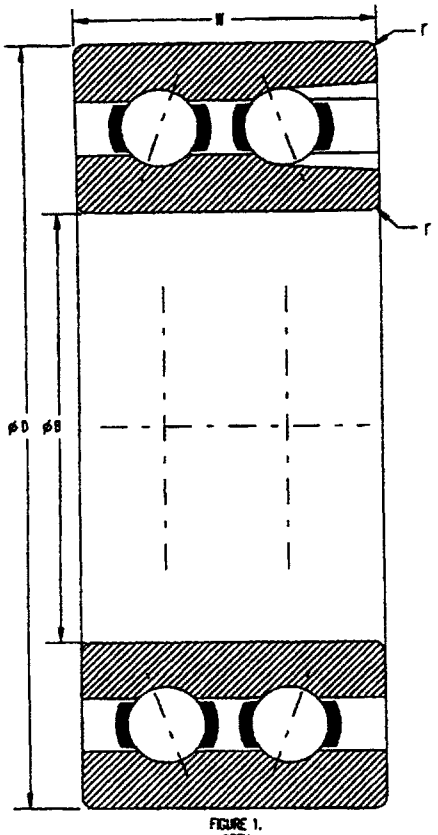


FIGURE 1.
OPEN
(CODE A)

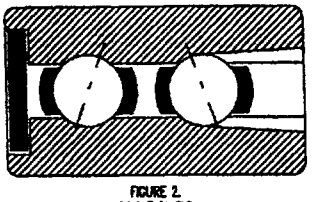


FIGURE 2.
SINGLE SHIELD
OPPOSITE FILLING SLOT SIDE
(NORMAL POSITION)
(CODE B)

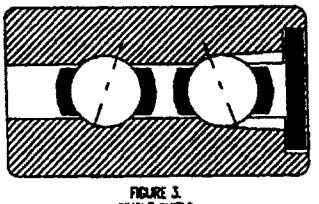


FIGURE 3.
SINGLE SHIELD
ON FILLING SLOT SIDE
(REVERSE POSITION)
(CODE C)

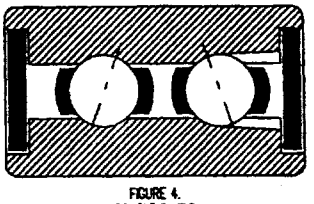


FIGURE 4.
DOUBLE SHIELD
(CODE D)

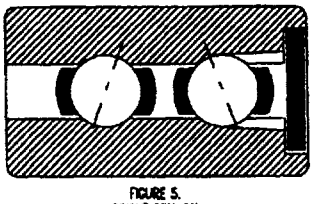


FIGURE 5.
SINGLE SEAL ON
FILLING SLOT SIDE
(REVERSE POSITION)
(CODE E)

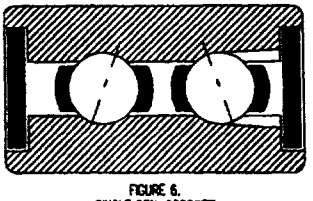


FIGURE 6.
SINGLE SEAL OPPOSITE
FILLING SLOT SIDE
SINGLE SHIELD ON
FILLING SLOT SIDE
(CODE F)

BEARING CONFIGURATIONS

AMSC/NA FSC 3110

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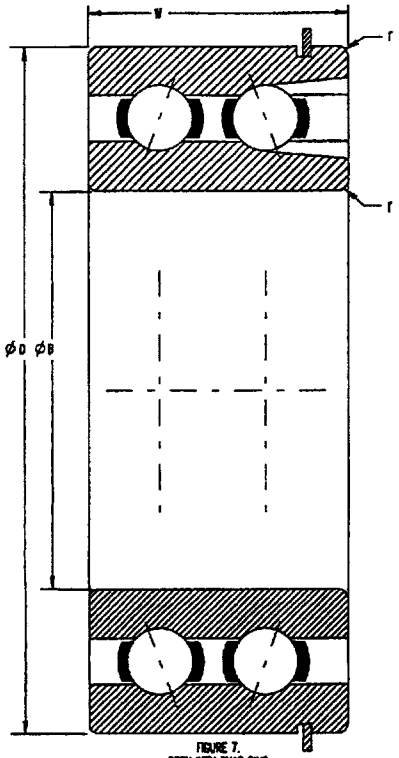


FIGURE 7.
OPEN WITH SNAP RING
ON FILLING SLOT SIDE
(NORMAL POSITION)
(CODE G)

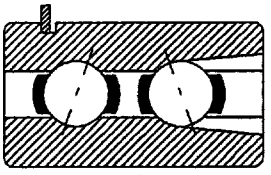


FIGURE 8.
OPEN WITH SNAP RING
OPPOSITE FILLING SLOT SIDE
(REVERSE POSITION)
(CODE H)

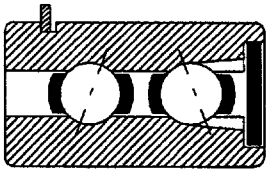


FIGURE 9.
SINGLE SHIELD,
REVERSE POSITION,
WITH SNAP RING,
(REVERSE POSITION)
(CODE J)

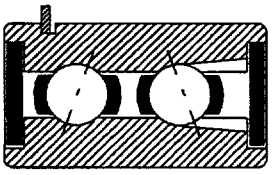


FIGURE 10.
DOUBLE SHIELD
WITH SNAP RING,
(REVERSE POSITION)
(CODE K)

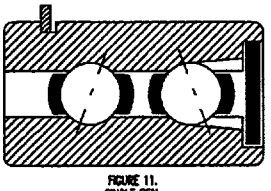


FIGURE 11.
SINGLE SEAL
(REVERSE POSITION)
WITH SNAP RING,
(REVERSE POSITION)
(CODE L)

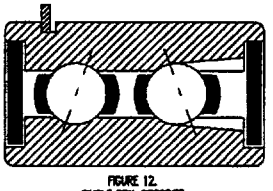


FIGURE 12.
SINGLE SEAL OPPOSITE
FILLING SLOT SIDE;
SINGLE SHIELD ON
FILLING SLOT SIDE;
WITH SNAP RING
(REVERSE POSITION)
(CODE M)

BEARING CONFIGURATIONS - (Continued)

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REQUIREMENTS:

1. Boundary dimensions. The boundary dimensions shall be specified in the part number by the size codes in Table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Shields, seals, and snap rings. Shields shall be fabricated from any material unless otherwise specified and shall be removable or nonremovable type. Snap rings shall be fabricated from ferrous material unless otherwise specified and shall have sufficient spring temper to set firmly in the ring grooves and shall be removable without the metal taking a permanent set. The snap ring and snap ring groove dimensions and their relative locations shall be in accordance with ANSI/AFBMA Standard 20. The number and positioning of shields, seals, and snap rings required shall be specified in the part number by the codes in table III.

4. Radial Internal Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

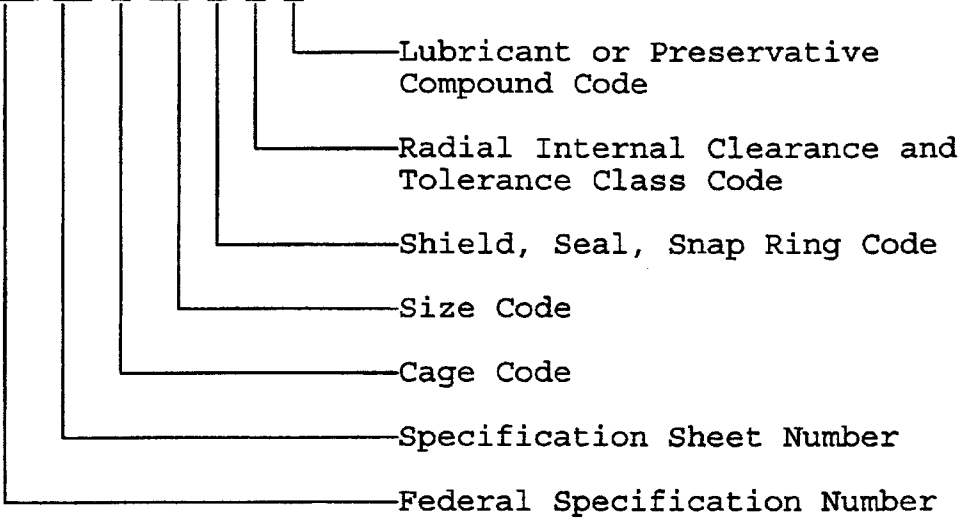
5. Tolerance Class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

6. Lubricant or Preservative Compound. Preservative compound shall be applied to all open bearings (no shields or seals) and to single closure bearings (single shield or single seal) that have no additional lubricant requirement. Thus, for these bearings the bearing part number shall include the "C" suffix. For double closure or single closure bearings that require lubricant the bearing part number shall include the "A" or "B" suffix. When grease is required, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25 to 50 percent filled. The lubricant or preservative compound shall be specified in the part number by the codes in table V.

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7. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields, seals, and snap rings, radial internal clearance and tolerance class, and lubricant or preservative compound.

EXAMPLE: FFB171/36 J 07 D 2 A



FFB171/36J07D2A indicates - Bearing, cage-pressed steel or molded plastic; bore 35mm; outside diameter 72mm; width 27mm; double shield; radial internal clearance symbol 0 (normal); tolerance class ABEC-1; grease in accordance with MIL-G-81322.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic load rating (pounds)
00	10	30	14.3	0.6	15	1635
01	12	32	15.9	0.6	17	1635
02	15	35	15.9	0.6	20	2155
03	17	40	17.5	0.6	22	2745
04	20	47	20.6	1.0	26	4000
05	25	52	20.6	1.0	31	3515
06	30	62	23.8	1.0	36	5910
07	35	72	27.0	1.1	42	7545
08	40	80	30.2	1.1	47	8600
09	45	85	30.2	1.1	52	9335
10	50	90	30.2	1.1	57	10120
11	55	100	33.3	1.5	63.5	11815
12	60	110	36.5	1.5	68.5	14650
13	65	120	38.1	1.5	73.5	16090
14	70	125	39.7	1.5	78.5	17640
15	75	130	41.3	1.5	84	18250
16	80	140	44.4	2.0	90	23680
17	85	150	49.2	2.0	95	24480
18	90	160	52.4	2.0	100	29510
19	95	170	55.6	2.1	107	36985
20	100	180	60.3	2.1	112	39160
21	105	190	65.1	2.1	117	41375
22	110	200	69.8	2.1	122	45090

1/ Size codes for Table I are industry bore code numbers.
2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.
3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)
Y	Pressed brass

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TABLE III. Codes for Shields and Seals.

Code	Shield and snap ring configuration
A	open (no shields) - fig.1
B	single shield opposite filling slot side, (normal position) - fig. 2
C	single shield on filling slot side, (reverse position) - fig. 3
D	double shield - fig. 4
E	open with snap ring on filling slot side, (normal position) - fig. 5
F	open with snap ring opposite filling slot side, (reverse position) - fig. 6
G	single shield, normal position, with snap ring, (normal position) - fig. 7
H	single shield, normal position, with snap ring, (reverse position) - fig. 8
J	single shield, reverse position, with snap ring, (normal position) - fig. 9
K	single shield, reverse position, with snap ring, (reverse position) - figure 10
L	double shield with snap ring, (normal position)-fig. 11
M	double shield with snap ring, (reverse position)-fig. 12

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (less than normal)	ABEC-1
2	Symbol 0 (normal)	
3	Symbol 3 (greater than normal)	
4	Symbol 4 (greater than symbol 3)	
5	Symbol 2 (less than normal)	ABEC-5
6	Symbol 0 (normal)	
7	Symbol 3 (greater than normal)	
8	Symbol 4 (greater than symbol 3)	

TABLE V. Codes for Lubricant and Preservative Compound.

Code	Lubricant or Preservative Compound
A	Grease in accordance with MIL-G-81322
B	Grease in accordance with DOD-G-24508
C	Preservative compound in accordance with MIL-P-197

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MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - OS
DLA - IS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

PREPARING ACTIVITY:
DLA-IS

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